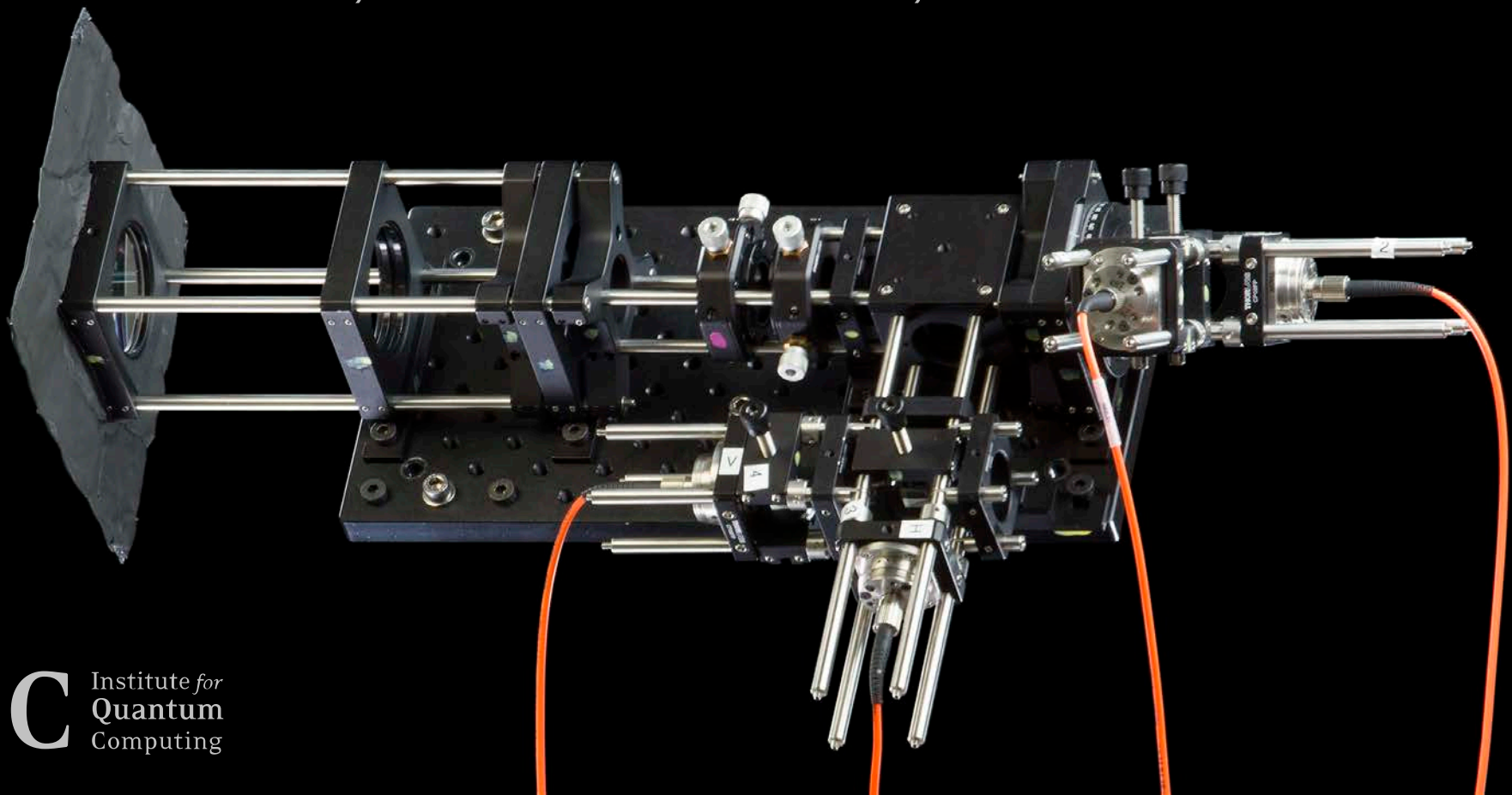
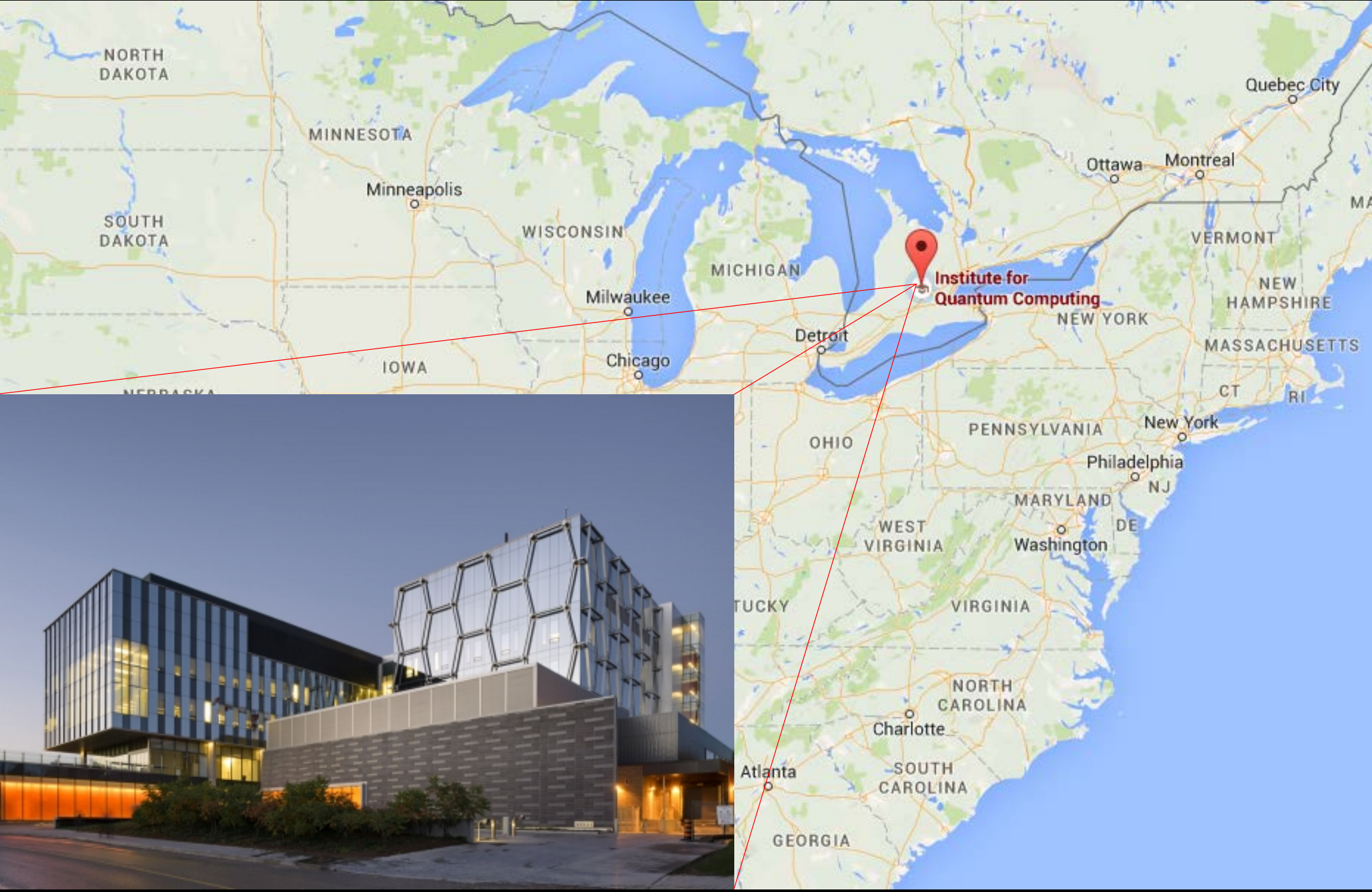
QCrypt 2015, Tokyo, 28 September – 3 October 2015

Spatial-mode detector efficiency mismatch security loophole in free-space QKD

Poompong Chaiwongkhot, Shihan Sajeed, Jean-Philippe Bourgoin, Thomas Jennewein, Norbert Lütkenhaus, Vadim Makarov



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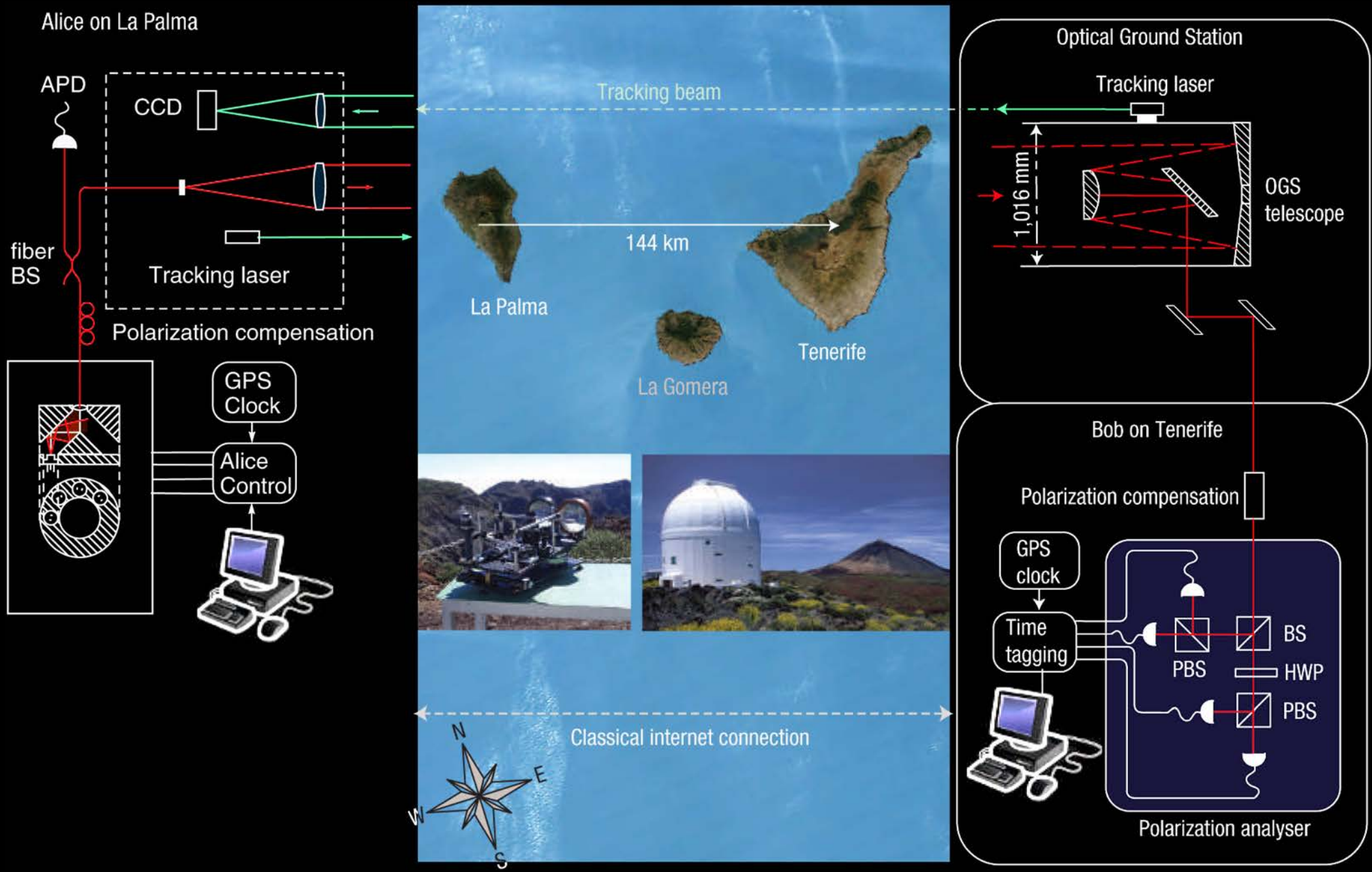
IQC Institute for Quantum Computing

https://uwaterloo.ca/institute-for-quantum-computing
Google map

Outline

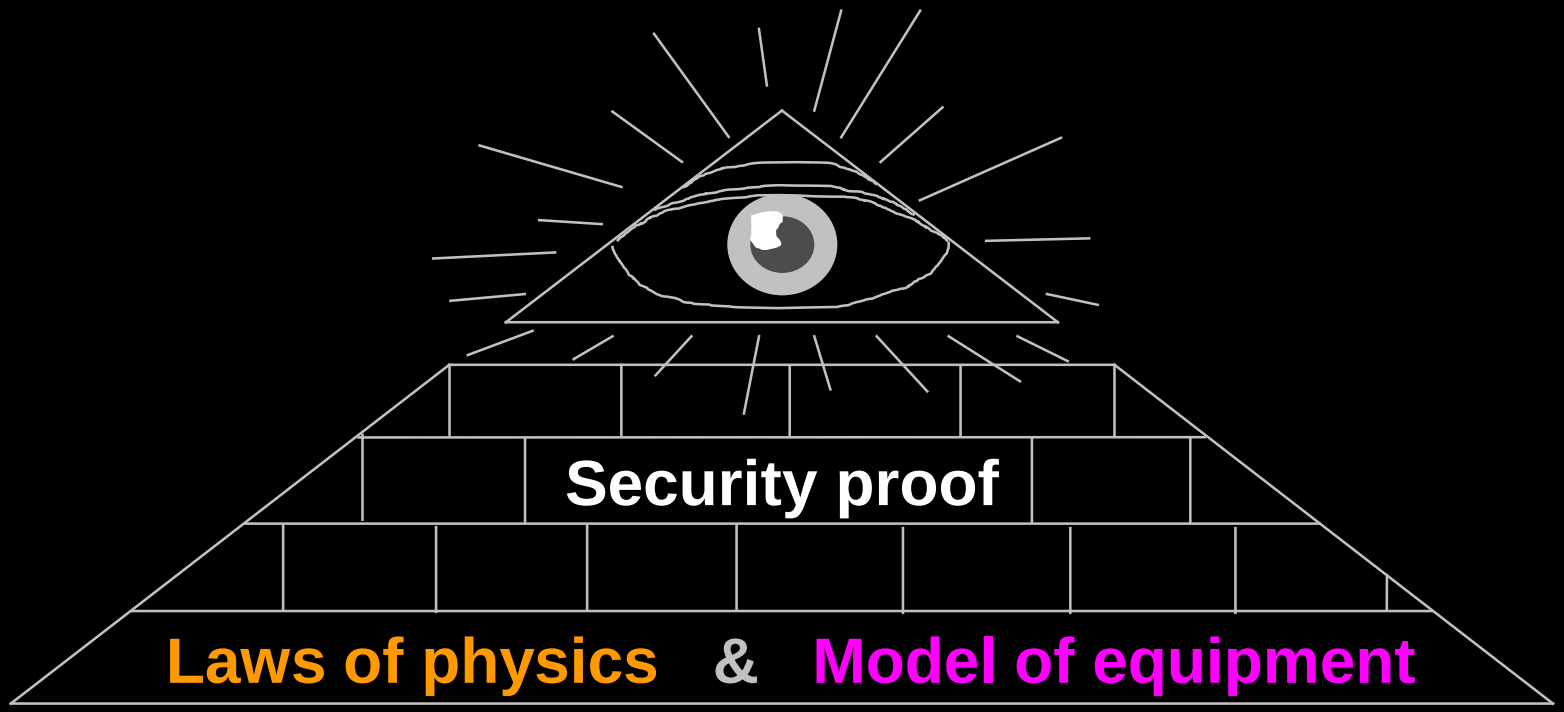
- Free-space QKD
- Security verification of QKD system
- Spatial mode detection efficiency mismatch
  - System under test
  - Experiment
  - Countermeasure
  - Relate works

Free-space QKD



T. Schmitt-Manderbach *et al.*, Phys. Rev. Lett. **98**, 010504 (2007)

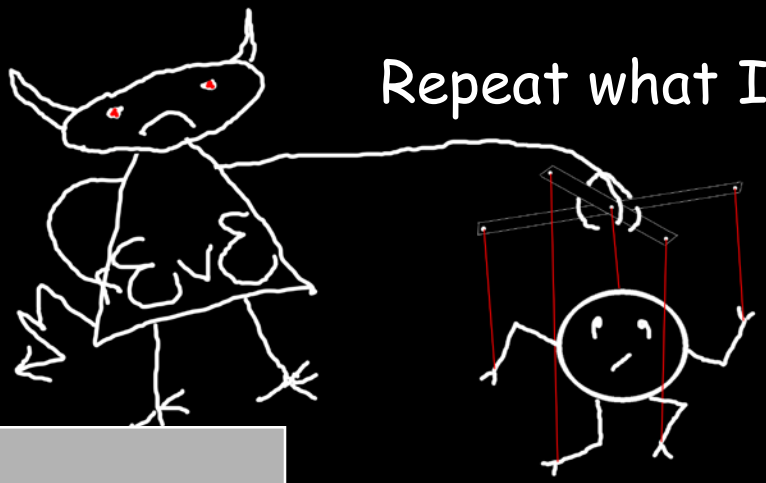
Security model of QKD

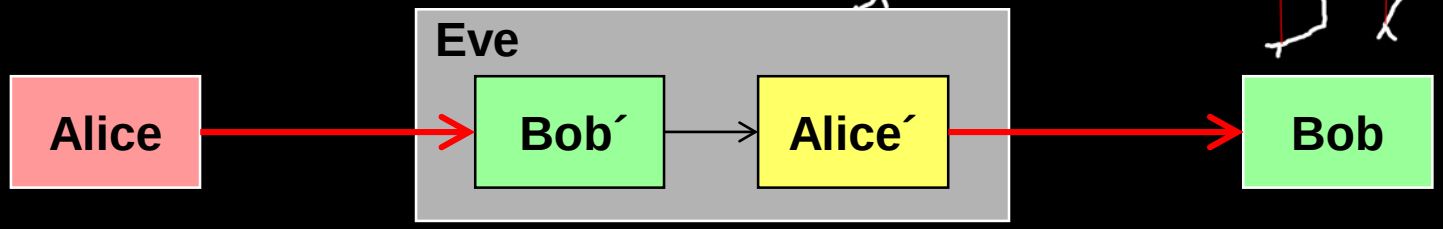


Hack

Integrate imperfection into security model

Intercept-resend attack with Bob's detectors controlled





intercept		resend		result	probability
H	1/2 H/V	H	H/V	H	1/2 η_H
			D/A ✗		
	1/2 D/A	D:A	H/V	H	~~1/4 η_H~~ 0
				V	~~1/4 η_V~~ 0
			D/A ✗		

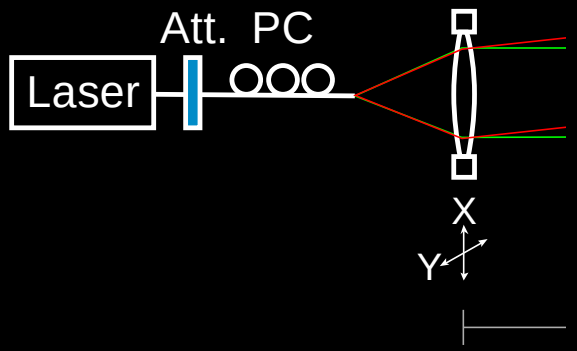
0 Error!

V. Makarov, D. R. Hjelme, J. Mod. Opt. **52**, 691 (2005)

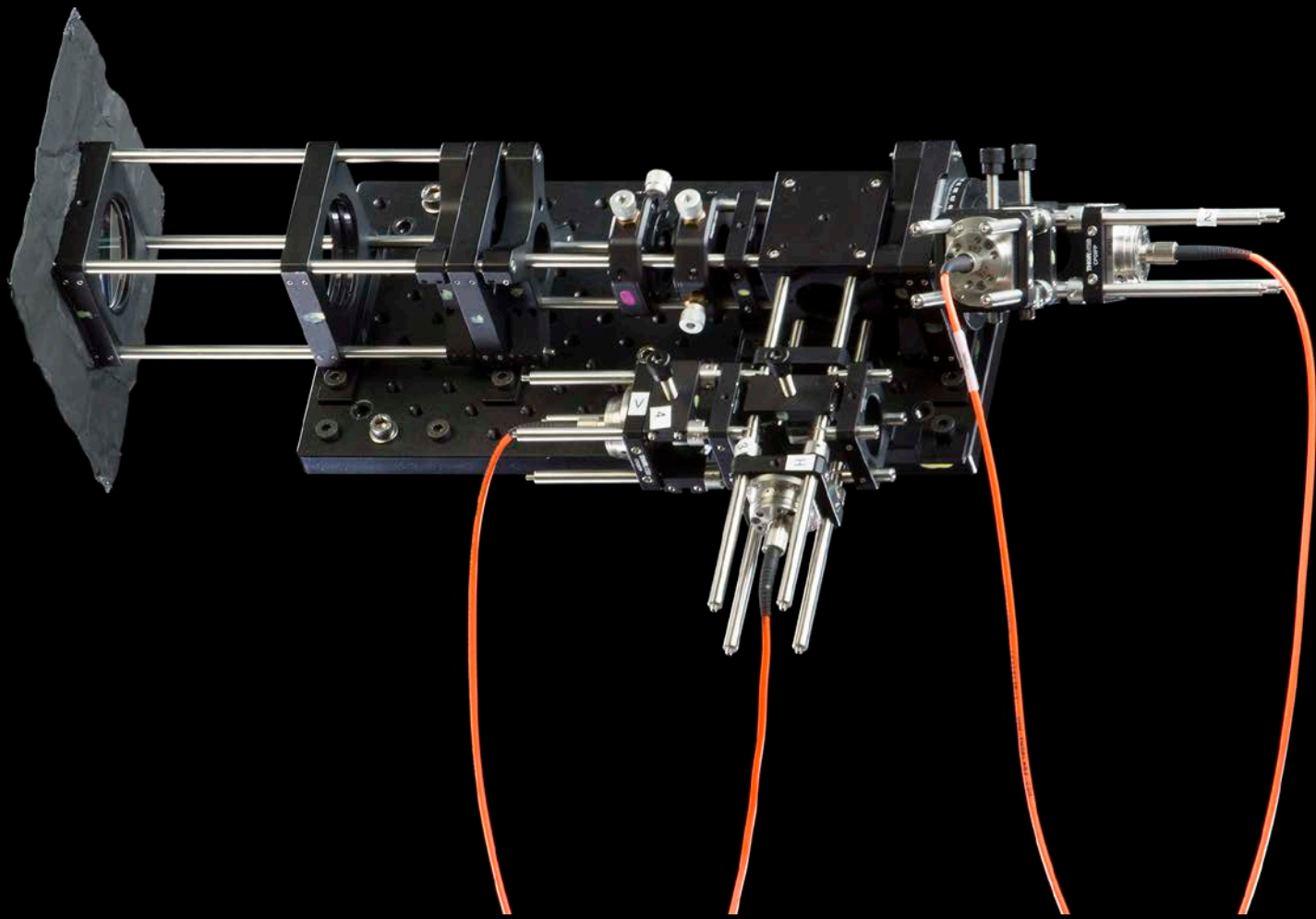
Efficiency mismatch in QKD receiver

Eve

Scanning

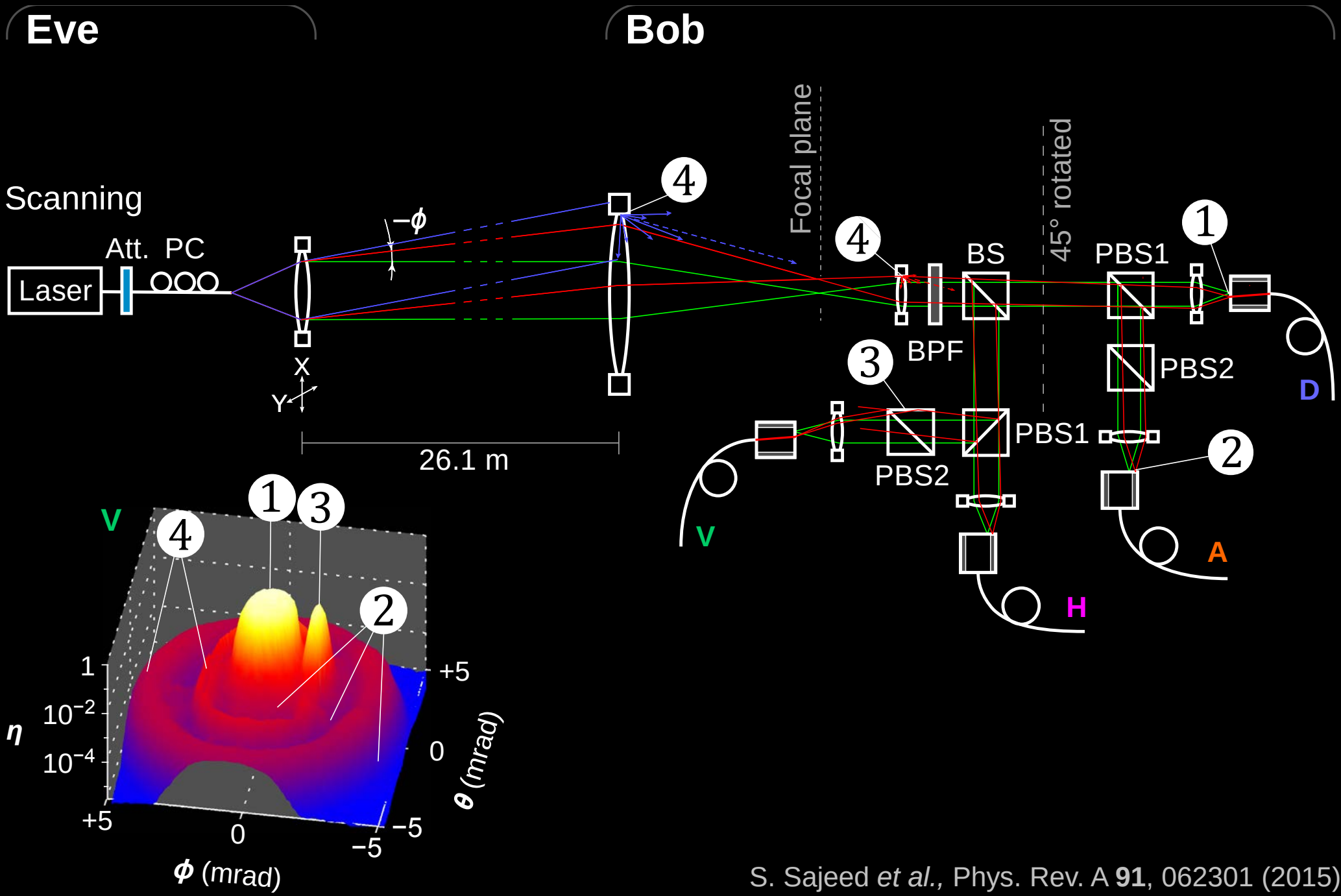


Bob



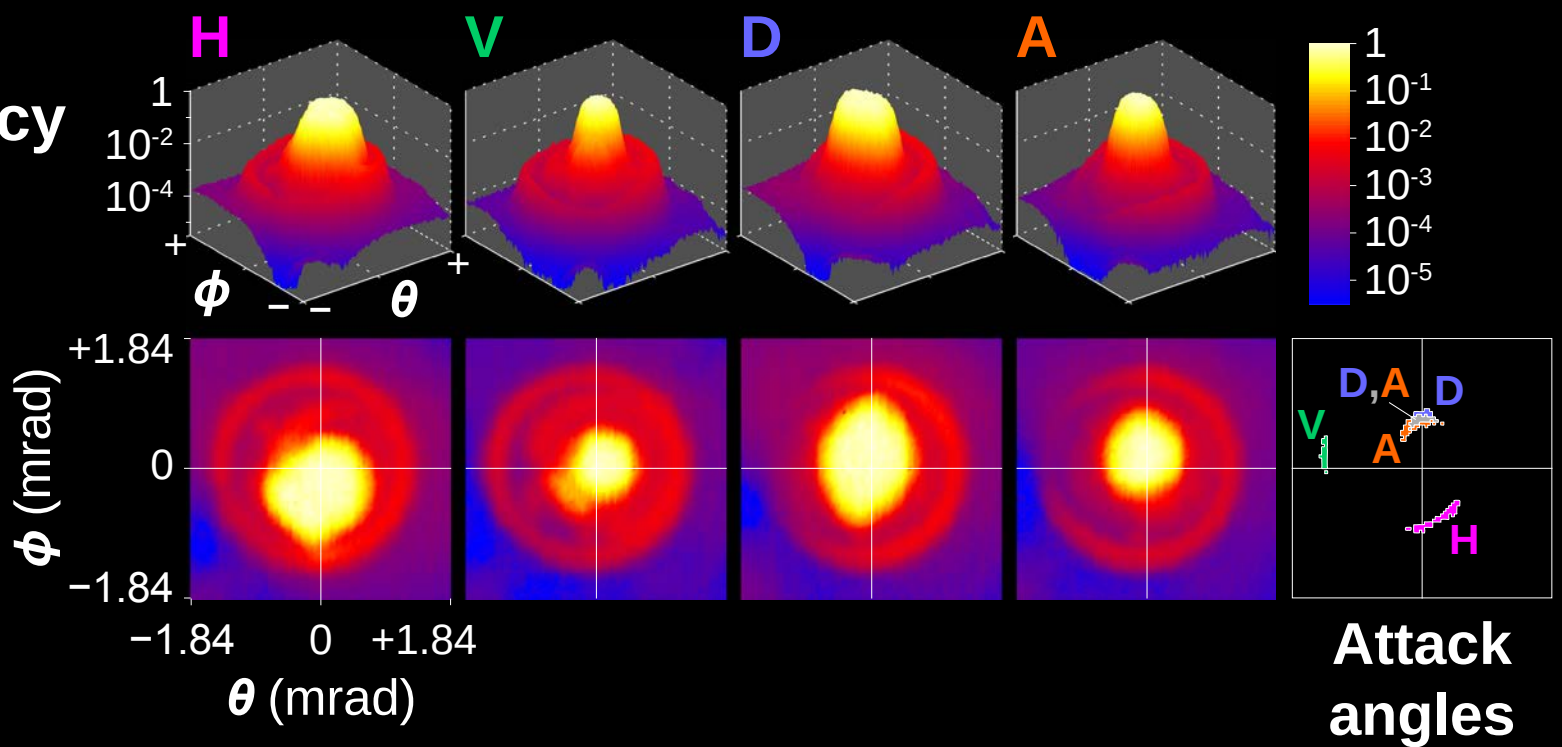
S. Sajeed *et al.*, Phys. Rev. A **91**, 062301 (2015)

Efficiency mismatch in QKD receiver



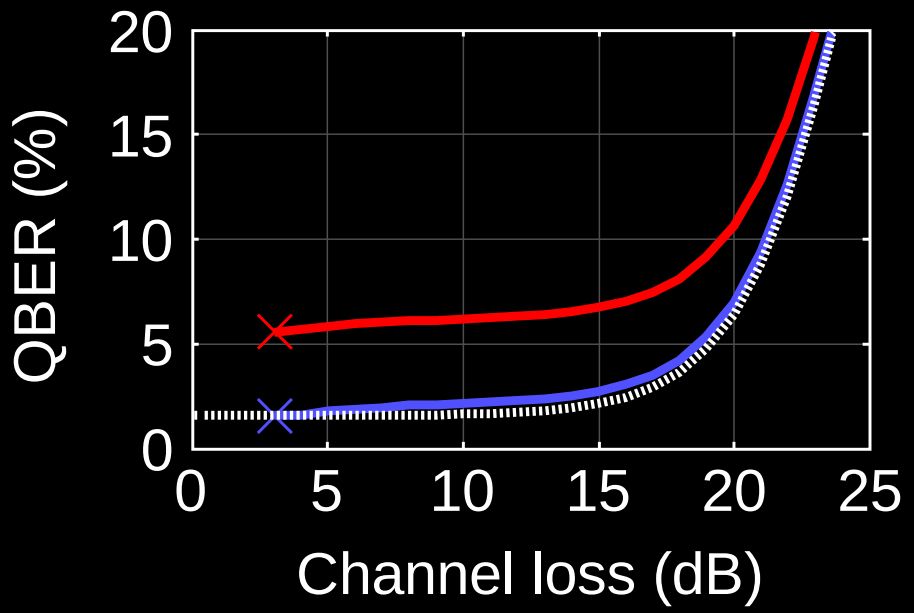
S. Sajeed *et al.*, Phys. Rev. A **91**, 062301 (2015)

Detector efficiency



Attack angles

Attack performance



Before attack

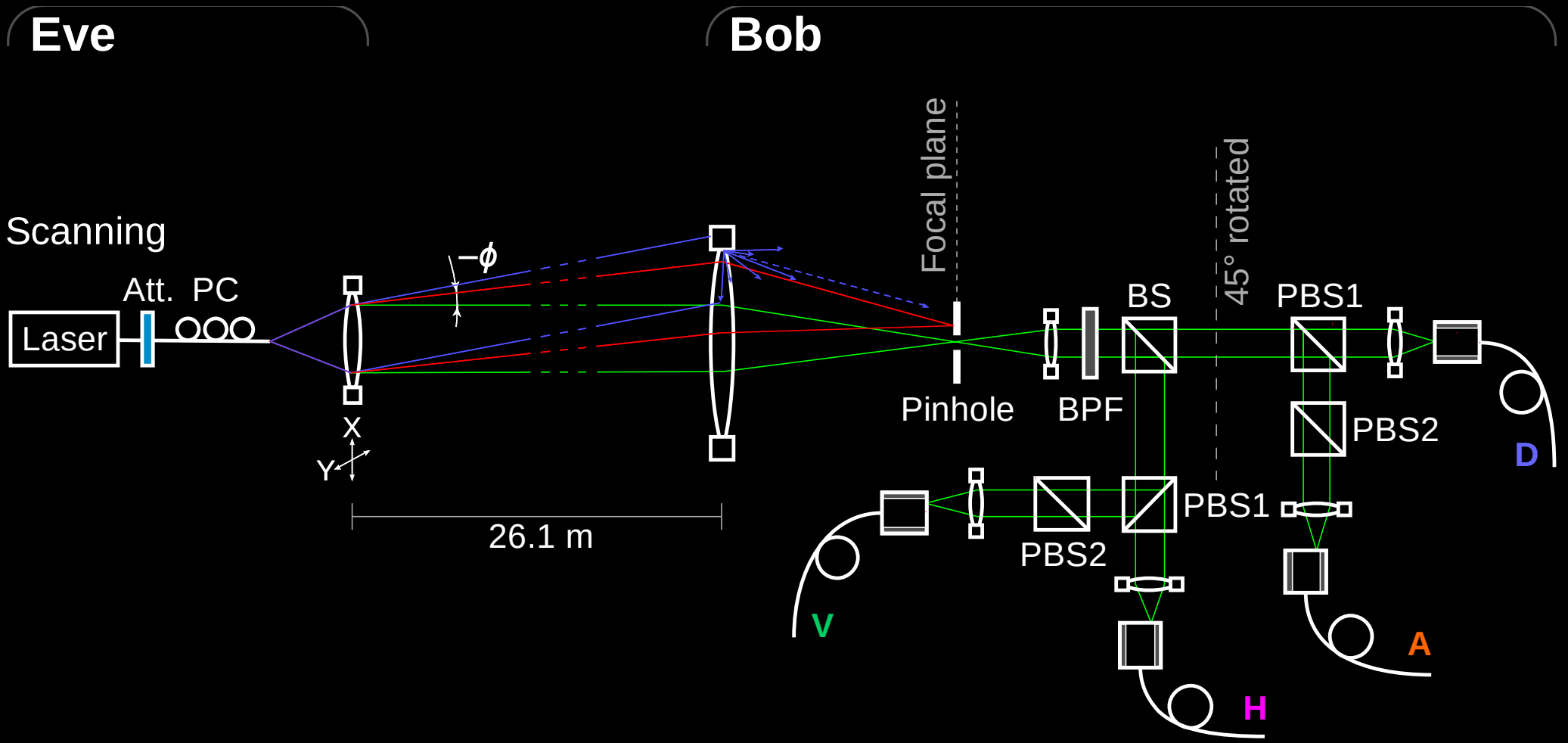
Under attack: match

total detection rate

+ equal channel rates

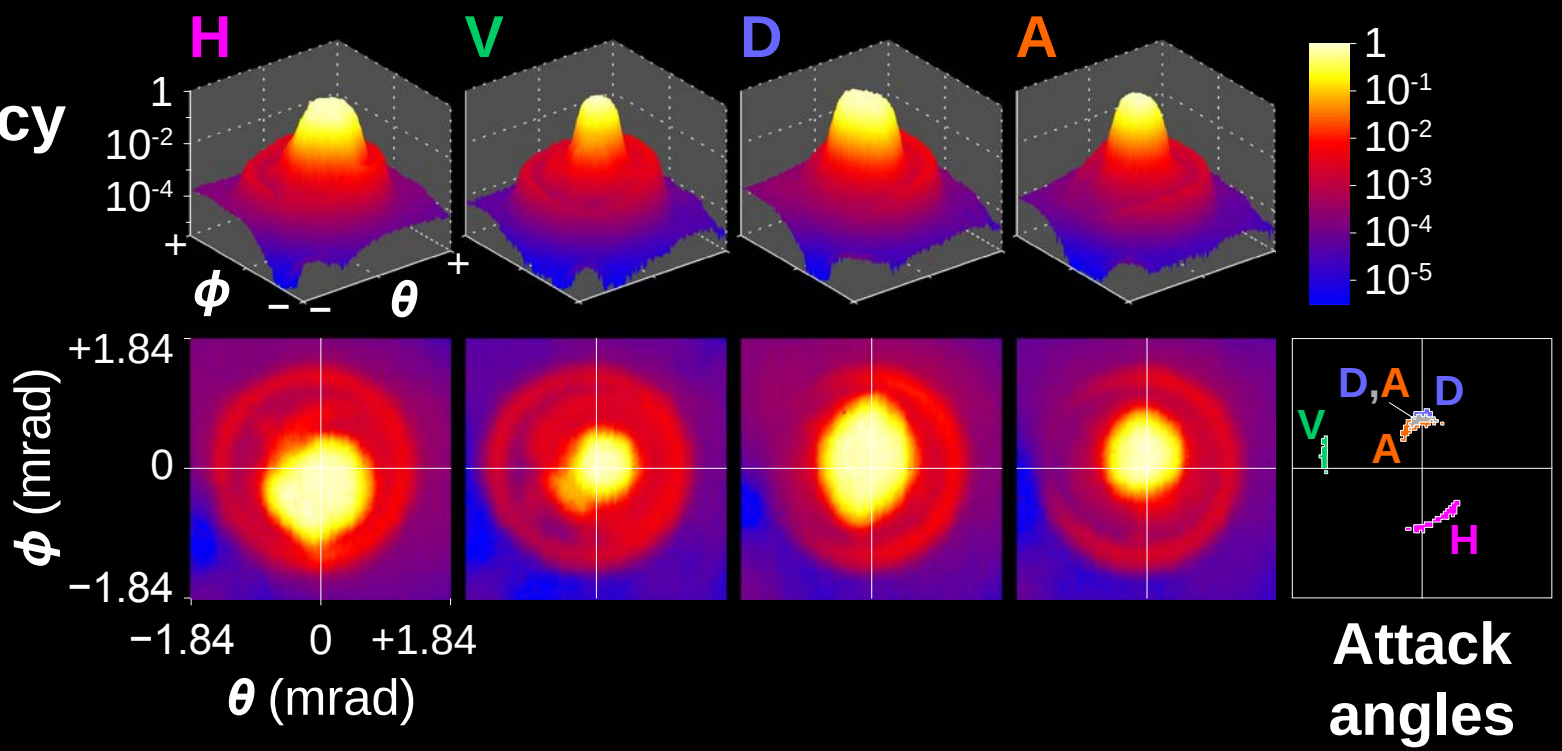
S. Sajeed *et al.*, Phys. Rev. A **91**, 062301 (2015)

Countermeasure



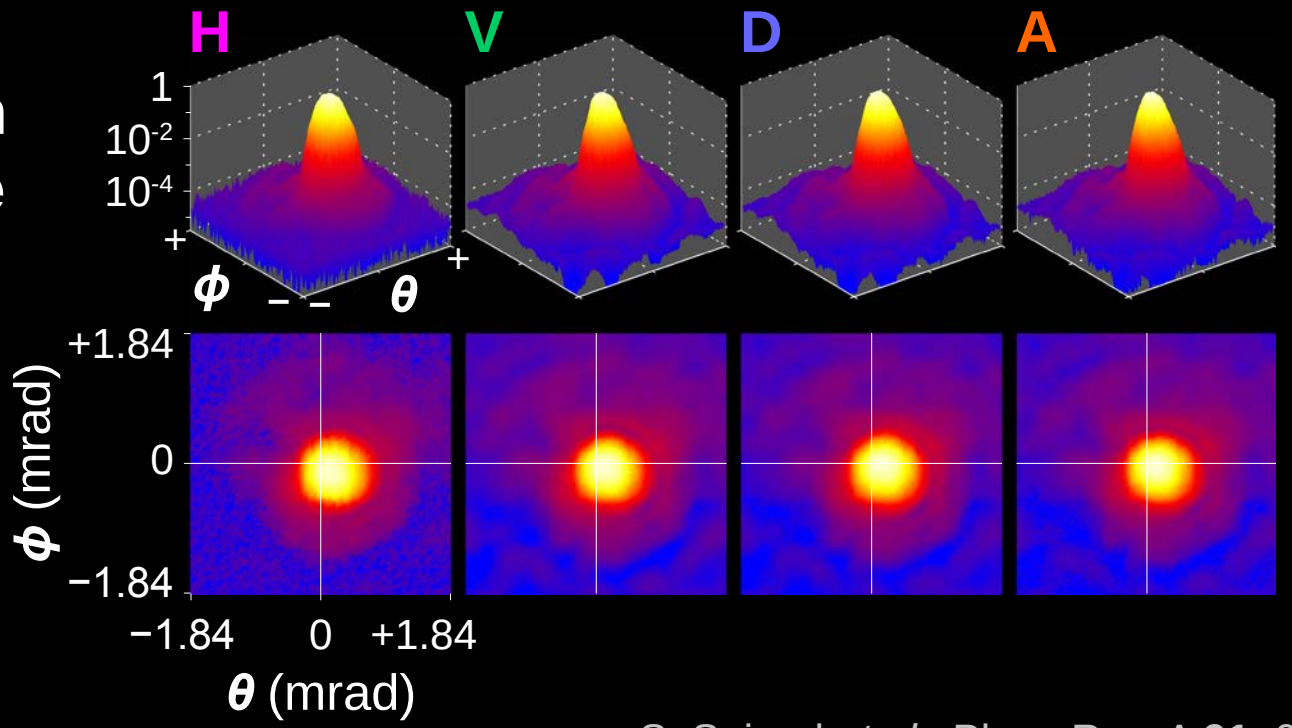
S. Sajeed *et al.*, Phys. Rev. A **91**, 062301 (2015)

Detector efficiency without pinhole



Attack angles

...and with 25 µm diameter pinhole

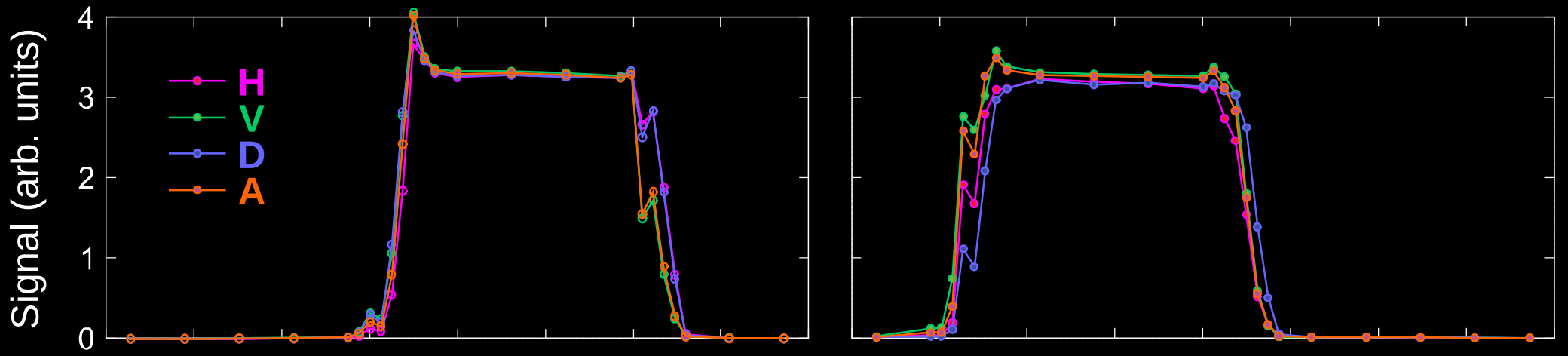


No attack found

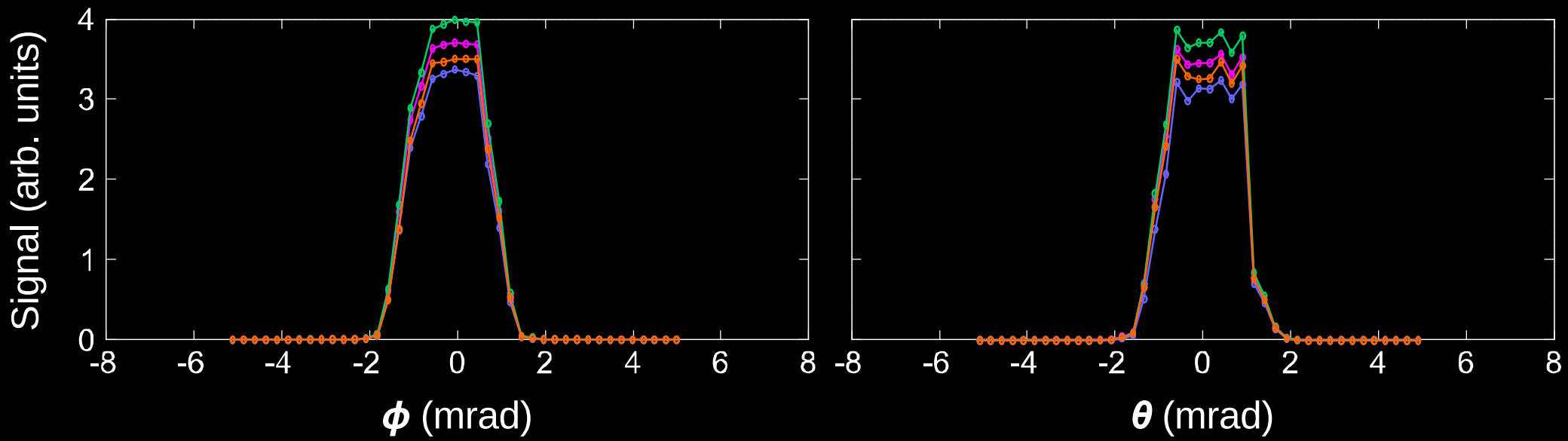
S. Sajeed *et al.*, Phys. Rev. A **91**, 062301 (2015)

Similar experiment by group in Munich

Without pinhole

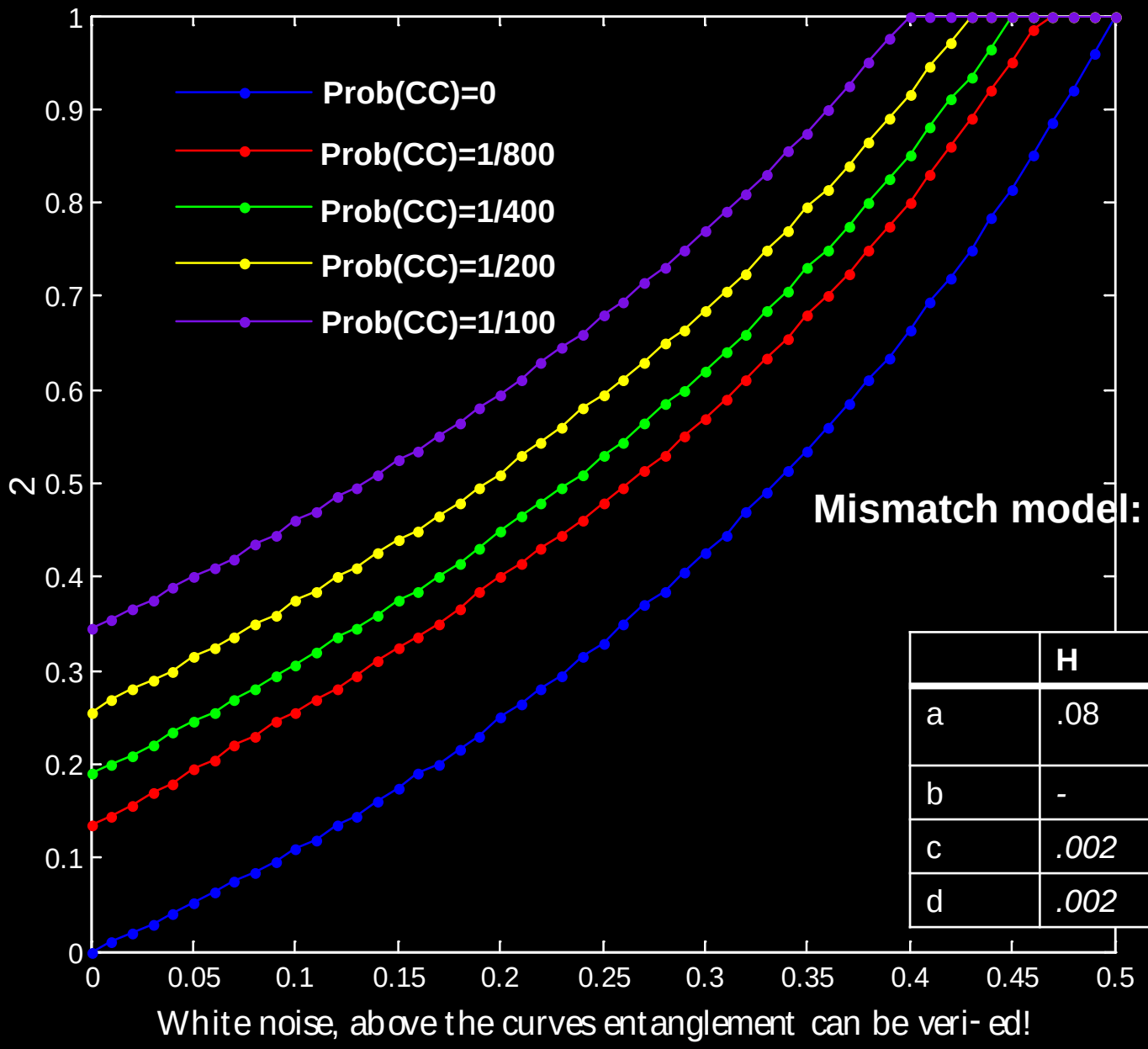


With pinhole



M. Rau *et al.*, IEEE J. Quantum Electron. **21**, 6600905 (2015)

Theoretical bound for efficiency mismatch



Mismatch model:

	H	V	D	A
a	1	η	η	η
b	η	1	η	η
c	η	η	1	η
d	η	η	η	1

	H	V	D	A
a	.08	-	*.00106*	*.00106*
b	-	.0008	*.0001*	*.0001*
c	*.002*	*.002*	.16	-
d	*.002*	*.002*	-	.04

Open questions & outlook

How robust is the pinhole countermeasure? Could a better attack be found? Can the system be proved secure?

Are there other countermeasures?

Will our attack still work under atmospheric turbulence?

Testing QKD and closing loopholes should eventually guarantee high level of security promised by theory.



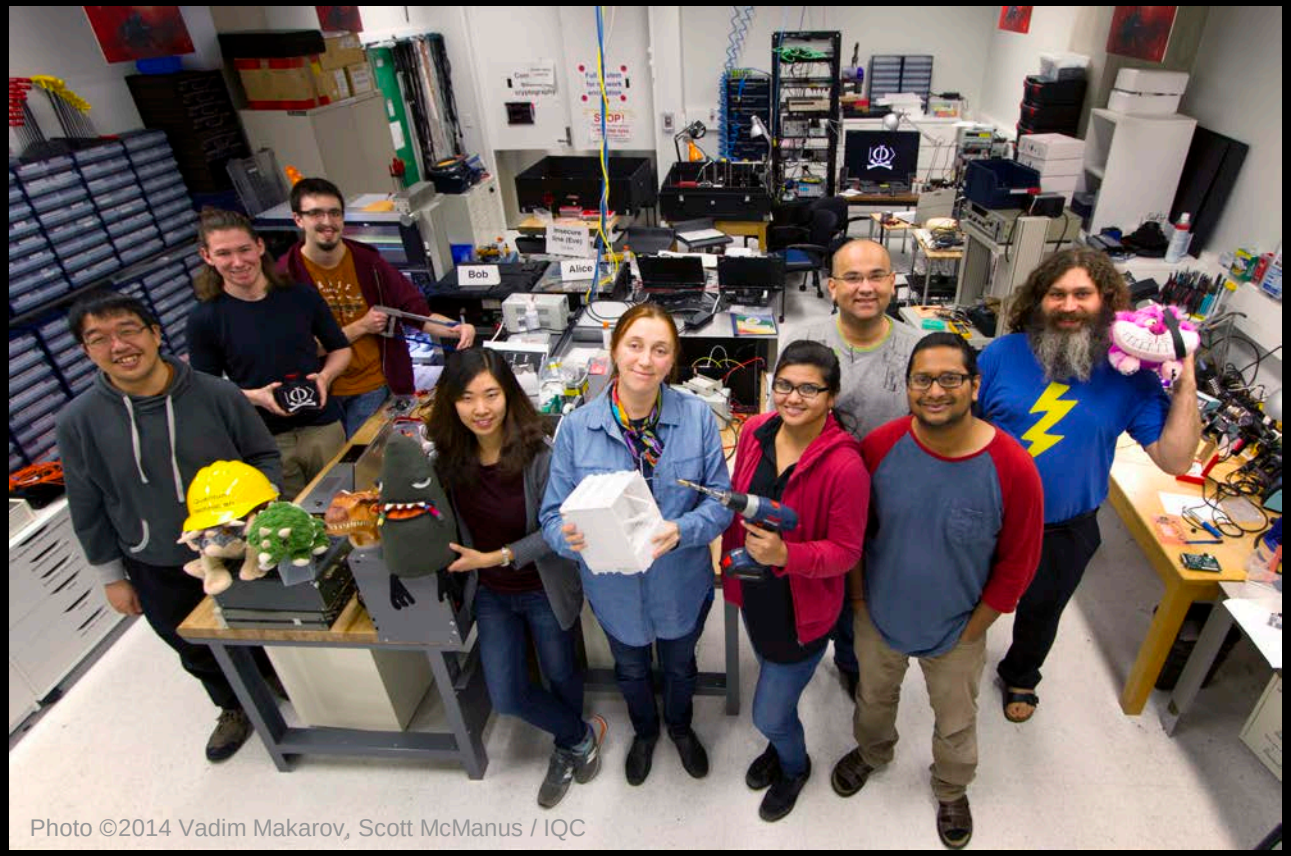


Photo ©2014 Vadim Makarov, Scott McManus / IQC



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CryptoWorks21









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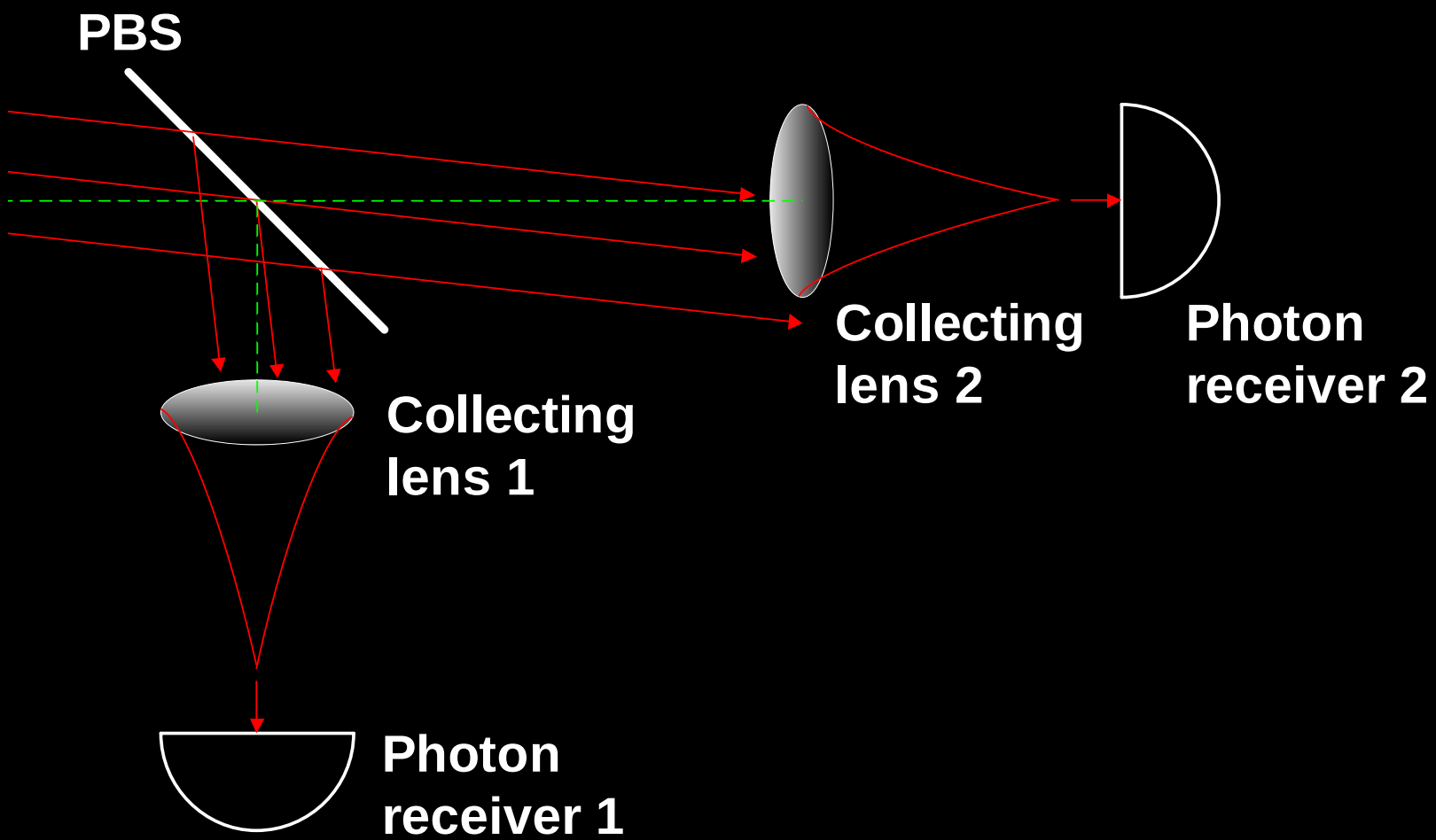
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Quantum hacking lab

www.vad1.com/lab

Optional slides

Example 3: academic. Efficiency mismatch



C.-H. F. Fung *et al.*, Quantum Inf. Comput. **9**, 0131 (2009)